Sample Space :

Toss a coin $\rightarrow$ $\Omega = \{\overset{\omega_1}{HH}, \overset{\omega_2}{HT}, \overset{\omega_3}{TH}, \overset{\omega_4}{TT}\}$

$P \rightarrow p_1, p_2 \; p_3 \; p_3$

$\sum p_i = 1$

$p_i \geq 0$

event $A = \{$At least 1 T appears$\}$

$= \{HT, TH, TT\}$

$P(A) = p_2 + p_3 + p_4 = 1 - p_1$

$= 1 - P(A')$

$\mathcal{F} = \{$subsets of $\Omega\} \rightarrow$ σ-field

A random variable X is a function $f : \Omega \rightarrow (-\infty, \infty)$

$X = \#$ of heads

$X(\omega_1) = 2$

$X(\omega_2) = 1$

$X(\omega_3) = 1$

$X(\omega_4) = 0$

$X_2 = 5X$

$X_3 = \#H's - \#T's$

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$X \sim Bin(n,p)$, X = H's in n tosses where $p = P(H)$

$P(X=k) = {}^nC_k \, p^k (1-p)^{n-k}$ for $k = 0, \dots, n$

$X \sim Bernoulli(p) \leftrightarrow X \sim Binomial(1,p)$

Y = longest run of H's

$X_1, X_2, \dots$ on Ω $= \{X_i : i \in \mathbb{N}\}$ → discrete

$\{X_t : t \geq 0\}$ → continuous

Stochastic Process → Sequence of these random variables

$F(x) = P(X \leq x)$ → Cumulative Distribution Function CDF

X_1, X_2 are independent if $P(X_1 \leq x_1, X_2 \leq x_2) = P(X_1 \leq x_1) . P(X_2 \leq x_2)$

$X_1, X_2, \dots X_n$ are mutually independent if

$P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = P(X_1 \leq x_1) P(X_2 \leq x_2) \dots P(X_n \leq x_n)$

for all $x_1, x_2 \dots x_n \in (-\infty, \infty)$

(then conversely true)

If X_i and X_j are independent for all i and j

Throw a fair dice twice.

X_1 = Sum of the outcomes is 7

X_2 = The outcome of first throw

X_3 = " " " second "

$\Omega = \{(i,j) : 1 \le i,j \le 6\}$

$$\overbrace{P(X_2 = i, X_3 = k)}^{1/36} = \underbrace{P(X_2 = i)}_{1/6} \cdot \underbrace{P(X_3 = k)}_{1/6}$$

X_2 and X_3 are independent

$$P(X_1 = j, X_2 = k) = \begin{cases} 0 & \text{otherwise} \\ \frac{1}{36} & X_3 = j-k \end{cases}$$

$$P(X_1 = 7, X_2 = 3, X_3 = 4) = \frac{1}{36}$$

$$\neq \underbrace{P(X_1 = 7)}_{\frac{1}{6}} \cdot \underbrace{P(X_2 = 3) \cdot P(X_3 = 4)}_{\frac{1}{36}}$$

Defn $X_1, X_2, \ldots$ is said to be independent sequence of r.v's if $P(X_1 \le x_1, \ldots X_n) = \prod_{i=1}^{n} P(X_i \le x_i)$ for all $-\infty \le x_1, x_2, \ldots x_n \le \infty$ and for all $n \ge 1$

Defn $X_1, X_2, \ldots$ is said to be an independent and identically distributed sequence (iid) of r.v's if

(i) $X_1, X_2, \ldots$ is an indep seq. of r.v's

(ii) for all i,j $P(X_i \le x) = P(X_j \le x)$ for all $x \in \mathbb{R}$

HW: i) X, Y, Z mutually independent r.v's with also identically distributed with $P(X=1)=\frac{1}{2}$, $P(X=-1)=\frac{1}{2}$. Consider the r.v's XY, YZ, XZ.

Q. Are they mutually independent?

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For a discrete r.v X: $E(X)=\sum_i x_i P(X=x_i)$

where $x_1, x_2, \dots$ are the values taken by X

$$Var(X)=E((X-E(X))^2)=E(X^2)-E(X)^2$$

For r.v's X, Y: $Cov(X,Y)=E((X-E(X))(Y-E(Y)))$

$$= E(XY)-E(Y)E(X)$$

HW: 2) $X_1, X_2, \dots, X_{n-1}$ iid Bernoulli (p)

$$X_n = \begin{cases} 1 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is odd} \\ 0 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is even} \end{cases}$$

Q. Is the sequence $X_1, \dots, X_n$ indep? [redacted]

Is the sequence $X_2, \dots X_n$ indep? [redacted]

Is the sequence $X_1, X_3, \dots X_n$ indep?

$\{a_0, a_1, \dots\}$ finite or infinite

The generating function of this sequence is defined as

$$A(s) = \sum_{i=0}^{\infty} a_i s^i = \lim_{n \to \infty} \underbrace{\sum_{i=0}^{n} a_i s^i}$$

polynomial of s in degree n

For $|s| < R$ where
R is Radius of Convergence

$$\{1, 1, 1, \dots\} \to A(x) = \sum_{i=0}^{\infty} x^i$$

$$= 1 + x + x^2 + x^3 + \dots$$

$$= \frac{1}{1-x} \quad \text{for } |x| < 1$$

Radius of convergence

X is a non negative integer valued rv
$X = k, \; k \geq 0$

$$P(X=k) = p_k \Rightarrow \left(\sum_{k=0}^{\infty} p_k = 1 \right)$$

$$P(s) = \sum_{k=0}^{\infty} p_k s^k \to \text{Probability Generating Function}$$

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Example: 1) $X \sim \text{Bernoulli}(p)$

$$X = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$$

$$P_X(s) = (1-p)s^0 + ps^1$$

2) $X \sim \text{Binomial}(n, p)$

$$P[X=k] = {}^nC_k\, p^k (1-p)^{n-k} \quad \text{for } k \in [0, n]$$

$$P_X(s) = \sum_{k=0}^{n} {}^nC_k\, p^k (1-p)^{n-k} s^k = \sum_{k=0}^{n} {}^nC_k (ps)^k (1-p)^{n-k}$$

Binomial Theorem: $(a+b)^n = \sum_{k=0}^{n} {}^nC_k\, a^k b^{n-k}$

$$\Rightarrow P_X(s) = (ps + 1 - p)^n$$

→ Same as Bernoulli with $n = 1$

3) $X \sim \text{Geometric}(p)$

X: Number of Failures till first Success

$X \in [0, \infty)$

$$P(X=k) = (1-p)^k p \quad \text{for } k = 0, 1, \ldots$$

$$P_X(s) = \sum_{k=0}^{\infty} (1-p)^k p\, s^k = p\left(\sum_{k=0}^{\infty} ((1-p)s)^k\right)$$

$$= \frac{p}{1 - s(1-p)}$$

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$$\boxed{\text{In general: } P_X(s) = \sum_{k=0}^{\infty} p_k s^k} \quad \text{for all } |s| \leq 1$$

how?

$P_X(1) = 1$

6) $X \sim \text{Poisson}(\lambda)$

$$P[X = k] = \frac{e^{-\lambda}\lambda^k}{k!} \quad \text{for } k \geq 0$$

$$e^x = \frac{x^0}{\underline{0}} + \frac{x^1}{\underline{1}} + \frac{x^2}{\underline{2}} + \frac{x^3}{\underline{3}} + \dots$$

$$P_X(s) = \sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} s^k = \sum_{k=0}^{\infty} \frac{e^{-\lambda}(\lambda s)^k}{k!}$$

$$= (e^{-\lambda})(e^{\lambda s})$$

$$P_X(s) = e^{-\lambda(1-s)}$$

$$E(s^X) = \sum_{k=0}^{\infty} P[X=k]\, s^k$$

for a r.v X non neg integer valued

$$P_X(s) = \sum_{k=0}^{\infty} p_k s^k \quad |s| \leq 1$$

$$= p_0 s^0 + p_1 s^1 + p_2 s^2 + \dots$$

$$\frac{dP_X(s)}{ds} = 0 + p_1 + 2p_2 s + 3p_3 s^2 + \dots$$

$$\frac{d}{ds} P_X(s)\Big|_{s=0} = p_1 = P[X=1]$$

$$\frac{d^2}{ds^2} P_X(s)\Big|_{s=0} = 2p_2 = \lfloor\underline{2}\; P[X=2]$$

$$\boxed{\frac{d^n}{ds^n} P_X(s)\Big|_{s=0} = \lfloor\underline{n}\; P[X=n]}$$

$$a = \{a_0, a_1, a_2, \dots\} \rightarrow A(s)$$
$$b = \{b_0, b_1, b_2, \dots\} \rightarrow B(s)$$

$$c_n = a_0 b_n + a_1 b_{n-1} + \dots a_{n-1} b_1 + a_n b_0$$
$$= \sum_{i=0}^{n} a_i b_{n-i}$$

$$c = \{c_0, c_1, \dots\} \longrightarrow C(s)$$

c is convolution of a and b

$$A(s)B(s) = (a_0 + a_1 s + a_2 s^2 + \dots)$$
$$\times (b_0 + b_1 s + b_2 s^2 + \dots)$$

$$= a_0 b_0 + s\overbrace{(a_1 b_0 + a_0 b_1)}^{c_1} + s^2\overbrace{(a_2 b_0 + a_1 b_1 + a_0 b_2)}^{c_2} + \dots$$

$$\Rightarrow A(s)B(s) = C(s)$$

Independent $\begin{cases} X \text{ taking values } 0, 1, 2, \dots \\ Y \text{ taking values } 0, 1, 2, \dots \end{cases}$

$$\Rightarrow P[X=k, Y=l] = P[X=k]\, P[Y=l] \text{ for all } k, l$$

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$$P(X+Y=j) = P[X=0, Y=j] + P[X=1, Y=j-1] + \dots + P[X=j, Y=0]$$
$$= P(X=0)P(Y=j) + P(X=1)P(Y=j-1) + \dots + P(X=j)P(Y=0)$$
$$= p_0^X p_j^Y + p_1^X p_{j-1}^Y + \dots\ p_j^X p_0^Y$$

$P_{X+Y}(s) = P_X(s) P_Y(s)$ for X, Y indep rvs

<u>Notation</u>: if the seq c is a convolution of the sequences a and b, then we write $c = a * b$

$P_W(s) = P_Z(s) \iff$ W and Z have same probability distribution i.e. W is same rv as Z

(Proof by taking derivatives, and obtaining same probability at all points by setting $s=0$)

$X_1, X_2, X_3, \dots X_n$ iid Bernoulli(p)

$P_{X_1}(s) = (1-p) + ps = P_{X_k}(s),\ k \in [1, n]$

$$P_{X_1+X_2+X_3+\dots+X_n}(s) = (1-p+ps)^n$$
$$= P_{Binomial(n,p)}(s)$$

$\Rightarrow (X_1 + X_2 + X_3 + \dots + X_n) \sim$ Binomial (n, p)

$$\left.\begin{aligned} X_1 &\sim \text{Poisson}(\lambda) \\ X_2 &\sim \text{Poisson}(\mu) \end{aligned}\right\} \text{ indep}$$

$$P_{X_1+X_2}(s) = P_{X_1}(s) \cdot P_{X_2}(s)$$

$$= e^{-\lambda(1-s)} \cdot e^{-\mu(1-s)} = e^{-(\lambda+\mu)(1-s)}$$

$$= P_{\text{Poisson}(\lambda+\mu)}$$

$$\therefore (X_1 + X_2) \sim \text{Poisson}(\lambda+\mu)$$

$$E(X^k) = m_k \rightarrow k^{th} \text{ moment}$$

$$\frac{d^k}{ds^k} P_X(s) = \sum_{i=0}^{\infty} i(i-1)\ldots(i-k+1) p_i s^{i-k}$$

$$\left.\frac{d}{ds} P_X(s)\right|_{s=1} = \sum_{i=0}^{\infty} i p_i = E(X)$$

$$\left.\frac{d^2}{ds^2} P_X(s)\right|_{s=1} = \sum_{i=1}^{\infty} i(i-1) p_i = E[X(X-1)]$$

$$\vdots$$

$$\left.\frac{d^k}{ds^k} P_X(s)\right|_{s=1} = \underbrace{E[X(X-1)(X-2)\ldots(X-k+1)]}_{k^{th} \text{ factorial moment of } X}$$

$$P_X'(1) = E[X]$$

$$P_X''(1) = E[X(X-1)] = E[X^2 - X] = E[X^2] - E[X]$$

$$\Rightarrow E[X^2] = P_X''(1) + P_X'(1)$$

If X and Y are 2 rv's $P(X=k \mid Y=l)$

$$= \frac{P(X=k, Y=l)}{P(Y=l)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

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$$Var[X] = E[X^2] - (E[X])^2$$
$$= P_X''(1) \cdot + P_X'(1) - (P'_X(1))^2$$

drunk

A person takes a step in unit time

$$X_k = \begin{cases} 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{2} \end{cases}$$

X_k : Action at $t=k$

$0 \to$ Stays, $1 \to$ moves forward

at time $= N$, he collapses

N is indep of steps

$$S_N = \sum_{i=1}^{N} X_i$$

how far he has moved

Theorem: Let $X_1, X_2, \ldots$ be iid rv's and N a rv independent of $X_1, X_2, \ldots$, Let $S_N = \sum_{i=1}^{N} X_i$

then $P_{S_N}(s) = P_N(P_{X_1}(s))$

Proof:
$$P_{S_N}(s) = \sum_{i=0}^{\infty} P(S_N = i)\, s^i$$
$$= \sum_{i=0}^{\infty} s^i \sum_{j=0}^{\infty} P(S_N = i \mid N=j)\, P(N=j)$$

$$\sum_{j=0}^{\infty} P(S_N=i, N=j) = P(S_N=i)$$

$$= \sum_{i=0}^{\infty} s^i \sum_{j=0}^{\infty} P(S_N=i, N=j)$$

$$= \sum_{i=0}^{\infty} s^i \sum_{j=0}^{\infty} P\left(\sum_{l=0}^{\infty} X_l = i \mid N=j\right)$$
$$= \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} s^i\, P(N=j)\, P\left(\sum_{l=0}^{\infty} X_l = i\right)$$

$$= \sum_{j=0}^{\infty} P(N=j) \sum_{i=0}^{\infty} s^i \, P\left(\sum_{l=0}^{j} X_l = i \right)$$

$$= \sum_{j=0}^{\infty} P(N=j) \, \underbrace{(P_{X_1}(s))^j}_{\to t^j}$$

$$= \sum_{j=0}^{\infty} P(N=j) \, t^j = P_N(t)$$

$$= P_N(P_X(s))$$

Example: $X_1, X_2, \ldots$ iid $\quad X_1 = \begin{cases} 0 & \text{wp } \frac{1}{2} \\ 1 & \text{wp } \frac{1}{2} \end{cases}$

$$P_{X_1}(s) = \frac{1}{2} + \frac{1}{2}s$$

$N \sim$ geometric$\left(\frac{2}{3}\right)$

$$P_N(s) = \frac{2}{3-s}$$

$$P_{S_N}(s) = P_N(P_{X_1}(s))$$

$$= P_N\left(\frac{1+s}{2}\right) = \frac{2}{3 - \left(\frac{1+s}{2}\right)} = \frac{4}{5-s}$$

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pgf : $P_X(s) = E(s^X)$

mgf : $M_X(s) = E(e^{sX}) = \sum_{i=0}^{\infty} e^{si} P[X=i]$

$$\frac{d}{ds} M_X(s)\Big|_{s=0} = \sum_{i=0}^{\infty} i P[X=i] = E[X]$$

$$\frac{d^2}{ds^2} M_X(s)\Big|_{s=0} = \sum_{i=0}^{\infty} i^2 P[X=i] = E[X^2]$$

$$\frac{d^k}{ds^k} M_X(s)\Big|_{s=0} = E[X^k]$$

BRANCHING PROCESS

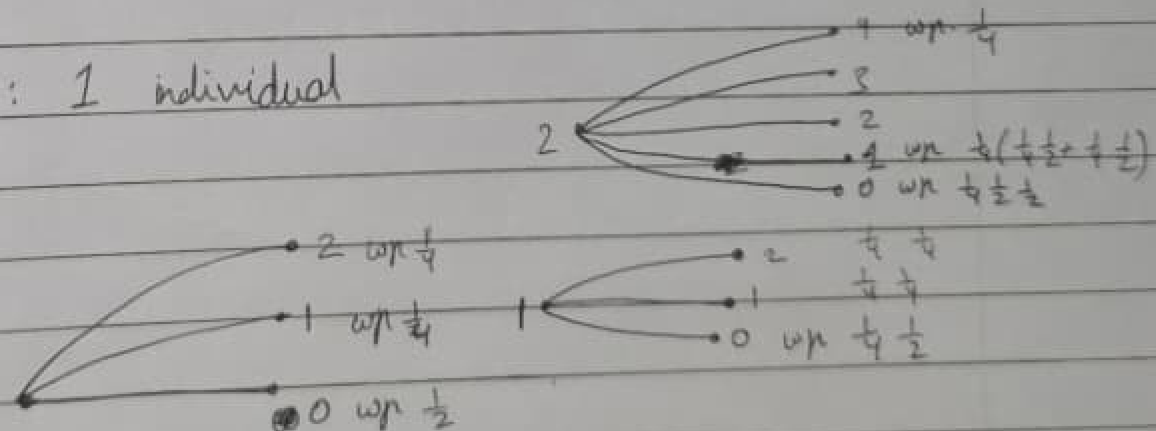
↳ galton-watson process

# of children of an individual is 0 w.p. $\frac{1}{2}$

1 w.p $\frac{1}{4}$

2 w.p $\frac{1}{4}$

gen 0 : 1 individual



H·W·1

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1) Let X, Y, Z be 3 mutually independent random variables and assume that they are identically distributed with $P(X=1)=\frac{1}{2}$, $P(X=-1)=\frac{1}{2}$. Consider the random variables XY, YZ, ZX. Are they independent?

	Case (X,Y,Z)	Probability of this case	XY	YZ	ZX
1	(1, 1, 1)	$P(X=1,Y=1,Z=1)=\frac{1}{8}$	1	1	1
2	(1, 1, -1)	$\frac{1}{8}$	1	-1	-1
3	(1, -1, 1)	$\frac{1}{8}$	-1	-1	1
4	(1, -1, -1)	$\frac{1}{8}$	-1	1	-1
5	(-1, 1, 1)	$\frac{1}{8}$	-1	1	-1
6	(-1, 1, -1)	$\frac{1}{8}$	-1	-1	1
7	(-1, -1, 1)	$\frac{1}{8}$	1	-1	-1
8	(-1, -1, -1)	$\frac{1}{8}$	1	1	1

For mutual independence, $P(XY=1, YZ=1, ZX=1) = \underbrace{P(X=1,Y=1,Z=1)+P(X=-1,Y=-1,Z=-1)}_{=\frac{1}{8}+\frac{1}{8}=\frac{1}{4}}$

$P(XY=1).P(YZ=1).P(ZX=1) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{8}$

$\therefore$ XY, YZ and ZX are not mutually independent

For pairwise independence:

Case 1: $XY=1$, $YZ=1$: $P(XY=1, YZ=1) = P(X=1,Y=1,Z=1)+P(X=-1,Y=-1,Z=-1)$

$= 2\left(\frac{1}{8}\right) = \frac{1}{4}$

$P(XY=1)\,P(YZ=1) = 4\left(\frac{1}{8}\right)\cdot 4\left(\frac{1}{8}\right) = \frac{1}{4}$

Case 2: $XY=1$, $YZ=-1$: $P(XY=1, YZ=-1) = P(X=1,Y=1,Z=-1)+P(X=-1,Y=-1,Z=1)$

$= 2\left(\frac{1}{8}\right) = \frac{1}{4}$

$P(XY=1)P(YZ=-1) = 4\left(\frac{1}{8}\right)\cdot 4\left(\frac{1}{8}\right) = \frac{1}{4}$

Case 3: $XY=-1$, $YZ=-1$: $P(XY=-1, YZ=-1) = P(X=1,Y=-1,Z=1)+P(X=-1,Y=1,Z=-1)$

$= 2\left(\frac{1}{8}\right) = \frac{1}{4}$

$P(XY=-1)P(YZ=-1) = 4\left(\frac{1}{8}\right)\cdot 4\left(\frac{1}{8}\right) = \frac{1}{4}$

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Case 4: $XY=-1, YZ=1$: $P(XY=-1, YZ=1) = P(X=1, Y=-1, Z=-1) + P(X=-1, Y=1, Z=1)$
$= 2\left(\frac{1}{8}\right) = \frac{1}{4}$
$P(XY=-1)\,P(YZ=1) = 4\left(\frac{1}{8}\right) \cdot 4\left(\frac{1}{8}\right) = \frac{1}{4}$

$\therefore$ By Symmetry, XY, YZ and ZX are pairwise independent.

2.) Let $X_1, X_2, \ldots, X_n$ be i.i.d Bernoulli (p) random variables and

$$X_n = \begin{cases} 1 & \text{if } \sum_{i=1}^{n-1} x_i \text{ is odd} \\ 0 & \text{if } \sum_{i=1}^{n-1} x_i \text{ is even} \end{cases}$$

Is the sequence independent? Justify.

Ans. Without Loss of Generality, let $n=3$, so X_1, X_2 are iid Bernoulli (p)
and $X_3 = \begin{cases} 1 & \text{if } X_1+X_2 \text{ is odd } (X_1=1, X_2=0 \text{ or } X_1=0, X_2=1) \\ 0 & \text{if } X_1+X_2 \text{ is even } (X_1=0, X_2=0 \text{ or } X_1=1, X_2=1) \end{cases}$

$P(X_1=0, X_2=0, X_3=1) = 0$ (As when $X_1=0, X_2=0$, then $X_3=0$
as $X_1+X_2=0$ is even)

$P(X_1=0) \cdot P(X_2=0) \cdot P(X_3=1) = (1-p)(1-p)\left(P[X_1=1, X_2=0] + P[X_1=0, X_2=1]\right)$

$= (1-p)^2\left(p(1-p) + p(1-p)\right)$
$= 2p(1-p)^3$

$\therefore P(X_1=0, X_2=0, X_3=1) = P(X_1=0) \cdot P(X_2=0) \cdot P(X_3=1)$ only if
$p=1$ or $p=1$ (not for any $p \in [0,1]$)

$\therefore X_1, X_2$ and X_3 are mutually dependent.

H·W· 2

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1) Let X be a non-negative integer valued random variable with probability generating function $P_X(s)$. Find the p.g.f of the random variables: (a) $Y = X+1$, (b) $Z = 2X$

Ans. In general, $P_X(s) = E[s^X]$

(a) $P_Y(s) = E[s^Y] = E[s^{X+1}]$

$$= E[s^X \cdot s]$$

$$= s\,E[s^X]$$

$$\Rightarrow P_Y(s) = s\,P_X(s)$$

(b) $P_Z(s) = E[s^Z] = E[s^{2X}]$

$$= E[(s^2)^X]$$

$$\Rightarrow P_Z(s) = P_X(s^2)$$

In General for X with p.g.f. $P_X(s)$ and $Y = aX+b$ where a, b are scalars, $\boxed{P_Y(s) = s^b P_X(s^a)}$

2) Let X be a non-negative integer valued random variable with probability generating function $P_X(s)$. For $n \geq 0$, let $a_n = P(X \leq n)$. Show that the generating function $A(s)$ of the sequence $\{a_n : n \geq 0\}$ is given by $A(s) = \dfrac{P_X(s)}{1-s}$.

Ans. For $n \geq 0$, $a_n = P(X \leq n)$. Let $A(s)$ be generating function of $\{a_n : n \geq 0\}$

$$A(s) = \sum_{i=0}^{\infty} a_i s^i = \sum_{i=0}^{\infty} P(X \leq i)\, s^i$$

$$= \sum_{i=0}^{\infty} s^i \left(\sum_{k=0}^{i} P(X \leq k) \right)$$

$$= s^0\big(P(X\leq 0)\big) + s^1\big(P(X\leq 0) + P(X \leq 1)\big) + s^2\big(P(X\leq 0) + P(X\leq 1) + P(X\leq 2)\big) + \cdots$$

$$= P(X\leq 0)(s^0 + s^1 + s^2 + \cdots) + P(X \leq 1)(s^1 + s^2 + \cdots) + \cdots$$

$$= \sum_{k=0}^{\infty} P(X=k) \sum_{i=k}^{\infty} s^i$$

$$= \left(\frac{1}{1-s}\right) \sum_{k=0}^{q} P(X=k) \cdot s^k$$

$$= \frac{P_X(s)}{1-s}$$

3) Let X be a non negative integer valued random variable with probability generating function : $P_X(s) = 0.16 + p^2 s + p s^2 + 0.6 s^3$

a) Find the value of p

b) What is $P(X \geq 1)$

c) What is $E(X)$

Ans. a) $P_X(1) = 1$

$\Rightarrow 0.16 + p^2 + p + 0.6 = 1$

$\Rightarrow p^2 + p - 0.24 = 0$

$\Rightarrow p = \dfrac{-1 \pm \sqrt{1 + 4(0.24)}}{2}$

$\Rightarrow p = \dfrac{-1 \pm \sqrt{1.96}}{2} = \dfrac{-1 \pm 1.4}{2}$

$= 0.2, -1.2$

An $P(X=2) = \dfrac{1}{\lfloor 2} \dfrac{d^2}{ds^2} P_X(s) \Big|_{s=0} = \dfrac{1}{\lfloor 2} 2p$

$= p$

$\Rightarrow \underline{p = 0.2}$ as $P(X=2) = p$ is non negative.

b) $P(X \geq 1) = 1 - P(X=0) = 1 - P_X(0) = 1 - 0.16$

$= 0.84$

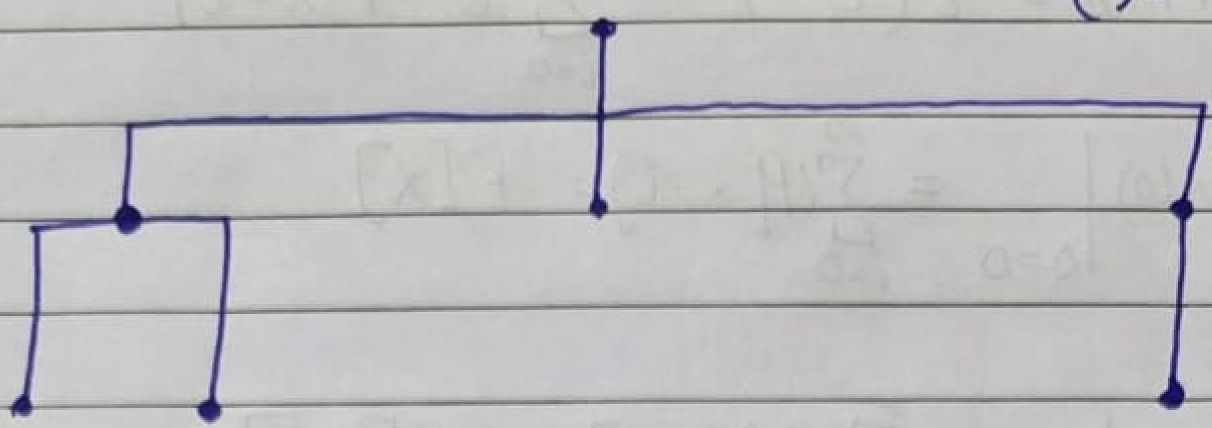
c) $E(X) = P'_X(1)$

$= p^2 + 2ps + 3(0.6)s^2 \Big|_{s=1}$

$= p^2 + 2p + 1.8 = 0.04 + 0.4 + 1.8$

$= 2.24$

Branching Processes



Assumptions:

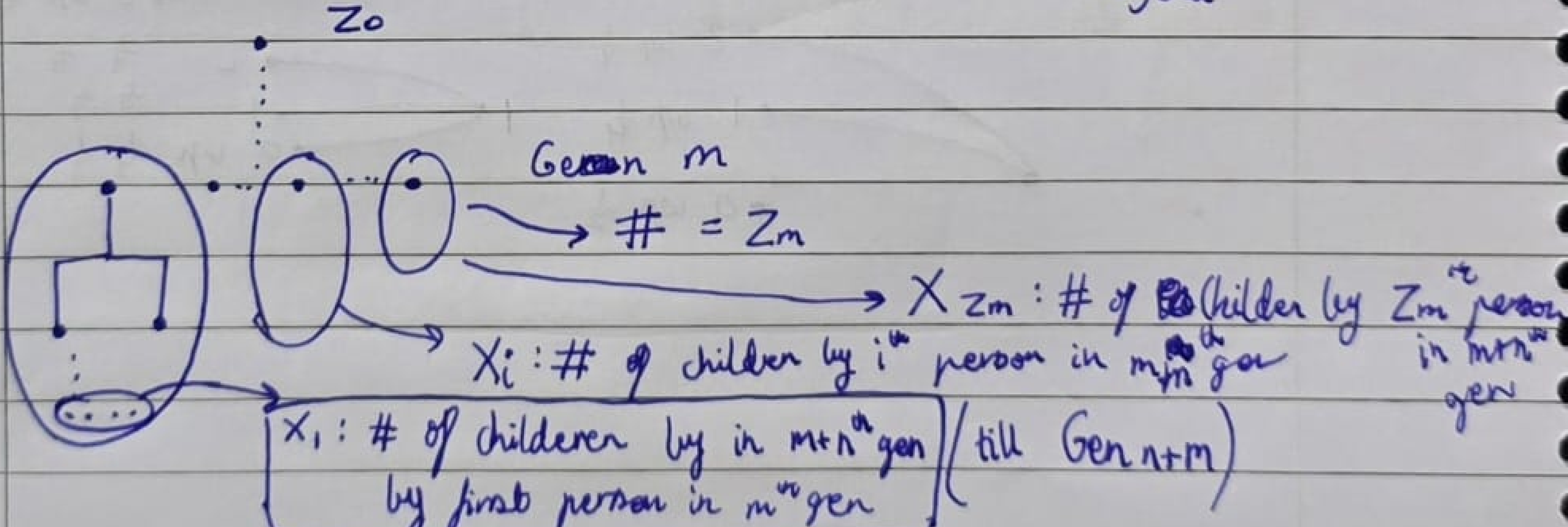
1) Two distinct individuals produce children: the no. of children produced by each of the individuals are independent of each other. (?mut or pairwise)
2) The no. of children produced by an individual is random with the same distribution as the no. of children produced by mother individual.

Notation: let Z_n = # of individuals in the n^{th} generation
$Z_0 = 1$

Let $G_n(s) = P_{Z_n}(s)$ be the p.g.f. of Z_n

Theorem: $G_{n+m}(s) = G_n(G_m(s))$

where $G_n(s) = \underbrace{G(G(G\cdots(G(s))))}_{n \text{ fold}}$



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$X_1, X_2, \ldots, X_{Z_m}$ are i.i.d. r.v. with X_i being the # individuals in the $(n+m)^{th}$ generation have an ancestor i in the m^{th} gen.

$$Z_{m+n} = X_1 + X_2 + \ldots + X_{Z_m}$$

$$G_{m+n}(s) = P_{Z_{m+n}}(s) = P_{Z_m}(P_{X_1}(s))$$
$$= G_m(G_n(s))$$

$G(s) =$ p.g.f. of Z_1
$G_n(s) =$ " " Z_n

→ Lemma: Let $\mu = E(Z_1)$, $\sigma^2 = Var(Z_1)$
Then $E(Z_n) = \mu^n$ and $Var(Z_n) = \begin{cases} n\sigma^2 & \text{if } \mu = 1 \\ \dfrac{\sigma^2(\mu^n - 1)\mu^{n-1}}{\mu - 1} & \text{if } \mu \neq 1 \end{cases}$

$$E[X] = \frac{d}{ds} P_X(s)\Big|_{s=1}, \quad \frac{d^2}{ds^2} P_X(s)\Big|_{s=1} = E[X(X-1)]$$

$$\frac{d}{ds} G_n(s) = \frac{d}{ds} G(G_{n-1}(s))$$
$$= G'(G_{n-1}(s)) . (G'_{n-1}(s))$$

$$, \frac{d}{ds} G_n(s)\Big|_{s=1} = G'(G_{n-1}(1)) . (G'_{n-1}(1))$$
$$= G'(1) . E(Z_{n-1})$$

$\Rightarrow E(Z_n) = \mu E(Z_{n-1}) \longrightarrow$ do induction

$$\frac{d^2}{ds^2} G_n(s)\Big|_{s=1} = \frac{d^2}{ds^2} G(G_{n-1}(s))\Big|_{s=1}$$

$$= \frac{d}{ds}\left(G'(G_{n-1}(s)) \cdot G'_{n-1}(s) \right)\Big|_{s=1}$$

$$\vdots$$

$$= G''(1)\left(G'_{n-1}(1)\right)^2 + G'(1)\, G''_{n-1}(1)$$

Example: $Z_1 \sim \text{Geometric}(p)$.

$P(Z_1 = k) = (1-p)p^k, \quad k \geq 0$

let $1-p = q$

$$G(s) = P_{Z_1}(s) = \frac{q}{1-ps}$$

$$G_2(s) = G(G(s)) = \frac{q}{1-pG(s)} = \frac{q}{1-p\left(\frac{q}{1-ps}\right)} = \frac{q(1-ps)}{1-ps-pq}$$

$$G_n(s) = \begin{cases} \dfrac{n-(n-1)s}{n+1-ns} & \text{if } p = q = \frac{1}{2} \\[2ex] \dfrac{q\left[p^n - q^n - ps\left(p^{n-1} - q^{n-1}\right)\right]}{p^{n+1} - q^{n+1} - ps\left(p^n - q^n\right)} & \text{if } p \neq q \end{cases}$$

Use induction

Using $P(X=0) = P_X(0)$ __/__/__

$$P(Z_n = 0) = \begin{cases} \frac{n}{n+1} & \text{if } p = q = \frac{1}{2} \\ \frac{q(p^n - q^n)}{p^{n+1} - q^{n+1}} & \text{if } p \neq q \end{cases}$$

$$\underbrace{\{Z_n = 0\}}_{E_n} \subseteq \underbrace{\{Z_{n+1} = 0\}}_{E_{n+1}}$$

$P(E_n) \longrightarrow P(\text{Ultimate extinction})$

$$\underset{\text{as } n \to \infty}{P(Z_n = 0)} = \begin{cases} 1 & \text{if } p \le q \quad p \le \frac{1}{2} \\ \frac{q}{p} & \text{if } p > q \quad p > \frac{1}{2} \end{cases} \qquad (p = 1?)$$

$n \to \infty$ exists (limit exists)

Theorem: Suppose $P(Z_n = 0) \to P(\text{ultimate extinction}) = \eta$
η is the smallest non negative root of the equation $s = G(s)$.
Also $\eta = 1$ if $\mu < 1$
$\eta < 1$ if $\mu > 1$
$\eta = 1$ if $\mu = 1$ and $\sigma^2 > 0$

(connected to whether Z_1 is a const. r.v. or not)

Proof: Let $\eta_n = P(Z_n = 0)$
$\eta_n \uparrow$, $\eta_n = G_n(0) = G(G_{n-1}(0))$
$= G(\eta_{n-1})$

$\eta = \lim_n \eta_n$

η $\Rightarrow \eta = G(\eta)$ (We need the fact that $G(s)$ is continuous)

η is the smallest non-neg solⁿ

Let ν be another solⁿ

$\eta_1 = G(0) \leq G(\nu)$

$\eta_2 = G_2(0) = G(G(0))$

$\Rightarrow \eta_2 \leq G(G(\nu))$

$\eta_n \leq G_{n-1}(\nu)$

$P_X(s) = p_0 + p_1 s + \dots + p_k s^k$
$\leq P_X(t)$ for $s \leq t$

$P_X(x)$ is a non decreasing function in s

As $n \to \infty$, $\eta_n \to \eta$

$G_{n-1}(\nu) \to \nu$

$\Rightarrow \eta < \nu$